

Answer: D

- 26** A source emits monochromatic light of frequency 5.0×10^{14} Hz at a rate of 0.10 W. The photons fall on the cathode of a photocell which gives a current of $80.0 \mu\text{A}$ in an external circuit. You may assume that this current consists of all the photoelectrons emitted. What is the percentage of the photons falling on the cathode which produce photoelectrons?

A 0.017%

B 0.17%

C 1.7%

D 100%

$$\text{No. of photons incident in 1 sec} = \frac{Pt}{hf} = \frac{(0.10)(1)}{(6.63 \times 10^{-34})(5.0 \times 10^{14})} = 3.017 \times 10^{17}$$

$$\text{No. of photoelectrons flowing in the circuit in 1 sec} = \frac{It}{e} = \frac{(80.0 \times 10^{-6})(1)}{(1.60 \times 10^{-19})} = 5.0 \times 10^{14}$$

$$\text{Percentage of photons which produce photoelectrons} = \frac{5.0 \times 10^{14}}{3.017 \times 10^{17}} = \mathbf{0.17\%}$$

N

Answer B